

12 Which suggested mechanism is consistent with the experimentally obtained rate equation?

	rate equation	suggested mechanism
A	$\text{rate} = k_1[\text{NO}]^2[\text{H}_2]$	$2\text{NO} + \text{H}_2 \xrightarrow{\text{slow}} \text{N}_2\text{O} + \text{H}_2\text{O}$ $\text{N}_2\text{O} + \text{H}_2 \xrightarrow{\text{fast}} \text{N}_2 + \text{H}_2\text{O}$
B	$\text{rate} = k_2[\text{H}_2][\text{I}_2]$	$\text{H}_2 \xrightarrow{\text{slow}} 2\text{H}$ $2\text{H} + \text{I}_2 \xrightarrow{\text{fast}} 2\text{HI}$
C	$\text{rate} = k_3[\text{HBr}]^2[\text{O}_2]$	$2\text{HBr} + \text{O}_2 \xrightarrow{\text{fast}} 2\text{HBrO}$ $\text{HBrO} + \text{HBr} \xrightarrow{\text{slow}} \text{H}_2\text{O} + \text{Br}_2$
D	$\text{rate} = k_4[\text{H}_2\text{O}_2][\text{I}^-]$	$\text{H}_2\text{O}_2 + \text{I}^- \xrightarrow{\text{fast}} \text{IO}^- + \text{H}_2\text{O}$ $\text{H}_2\text{O}_2 + \text{IO}^- \xrightarrow{\text{slow}} \text{I}^- + \text{H}_2\text{O} + \text{O}_2$